

SynthDoc - Project Report

Project Overview

Project: SynthDoc

Tech Stack: Java / Javalin 6.6.0

Purpose: AI document generator

Repository: github.com/malbinjvc/synth-doc

SynthDoc provides template-based document creation, AI content completion via Claude API, citation tracking with reference management, and multi-format export (text, markdown, HTML, JSON).

Architecture

- Template management (report, article, memo, etc.)
- Document CRUD with sections and metadata
- AI completion via Claude API (mocked for tests)
- Citation tracking with automatic references
- Export to text, markdown, HTML, JSON formats
- In-memory ConcurrentHashMap storage

API Endpoints

Endpoint	Method	Description
/api/templates	CRUD	Template management
/api/documents	CRUD	Document management
/api/documents/id/sections	POST	Add section
/api/documents/id/complete	POST	AI completion
/api/documents/id/citations	POST	Add citation
/api/documents/id/export	GET	Export document
/api/stats	GET	System statistics
/health	GET	Health check

Testing

Framework: JUnit 5

Tests: 117/117 passing (100%)

Security Audit

#	Area	Status
1	Hardcoded Credentials	PASS
2	Sensitive Files	PASS
3	SQL Injection	PASS
4	XSS/Validation	FAIL

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5	Authentication	FAIL
6	CORS/Headers	FAIL
7	File Upload	PASS
8	Docker Security	PASS
9	Infrastructure	PASS
10	Dependencies	PASS

Docker & CI/CD

- Multi-stage (temurin:21-jdk -> 21-jre-alpine), non-root
- GitHub Actions: build, test, Docker build with JDK 21